

OncoCountour

Cancer Data Visualization Application

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Home

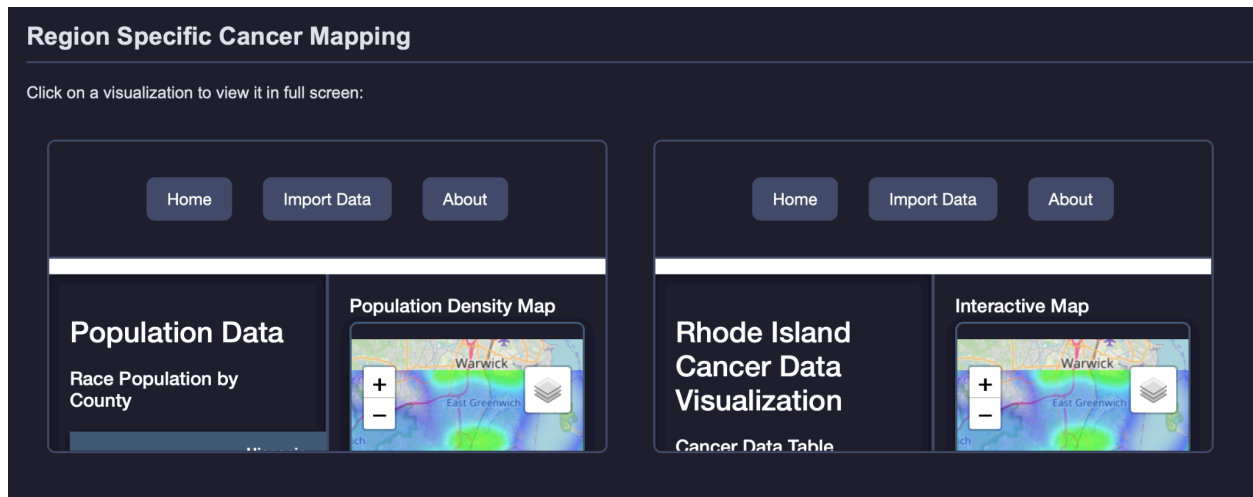
The Home Page serves as the main entry point to OncoContour. It provides quick access to the platform's key navigation and pre-built visualizations.

Navigation

Three navigation buttons are displayed at the top of every page throughout the application.

- Home: returns to the landing page
- Import Data: navigates to the data upload and visualization generation section
- About: provides information on the project, authors, and contact details

Pre-Built Visualizations



The Home Page features two preview panels showing off the graphing and mapping features applied to data and mock data specific to Rhode Island. Clicking either panel opens the full map.

- Population Distribution Map: contains population and demographic focused maps
- Cancer Incidence Map: has graphs detailing cancer types, timeframe, and location

These maps are interactive when opened in full screen, allowing users to zoom, pan, and hover over markers for detailed regional information.

Import Data

The Import Data page is the core workspace for loading your own datasets and generating custom visualizations. It accepts three types of CSV files, each serving a distinct role in the final output.

Overview

Once at least one valid file has been uploaded, users can click the Generate Visualization button to produce a custom interactive cancer map based on their data. Files are validated upon upload and will display either a green success message or a red error message with a description of the problem.

Note: All previously uploaded files are cleared each time a new session begins, so you will need to re-upload your data if you refresh or restart the application.

Data Import

Upload Your Data

Cancer Statistics Data

Upload CSV with format:
"City,State,CancerType1,CancerType2,...,Year1,Year2,..."

State must be 2-letter abbreviation (e.g., RI, MA)

Choose File

no file selected

Upload Cancer Data

Example Cancer Data Format

Your CSV should follow this pattern:

```
City,State,Bladder,Breast,Lung,Prostate,2015,2
Providence,RI,11,50,40,30,47,45,44,42,41,39,37
Lincoln,RI,32,20,15,10,18,35,16,15,14,13,12
Barrington,RI,33,25,18,12,23,55,21,20,19,18,17
```

County Race/Ethnicity Data

Upload CSV starting with "County" followed by any race or ethnicity you're tracking

Choose File

no file selected

Upload County Race Data

Example County Race Data Format

Your CSV should match this format:

```
County,White,Hispanic or Latino,Black,American
Bristol,45355,1943,773,105,1285,1,500,2774
Kent,147106,9665,3220,532,4882,40,3860,10723
Newport,72063,5592,2840,362,1570,64,2533,6211
Providence,402194,160323,53803,5362,28614,380,
Washington,116202,4578,1532,1024,2610,51,1992,
```

Age and Sex Population Data

Upload CSV starting with "Sex," followed by age ranges indicated by numbers

Choose File

no file selected

Upload Age/Sex Data

Example Age/Sex Data Format

Your CSV should follow this structure:

```
Sex,0-9,10-19,20-29,30-39,40-49,50-59,60-69,70
Male,55897,68369,78277,74753,63778,73199,67452
Female,54344,66057,75425,72340,62843,75137,736
```

Generate Visualization

File Types

There are three categories of data files that can be uploaded:

1. Cancer Statistics Data

This is the primary dataset required to generate a map. It contains city-level cancer incidence figures organized by cancer type and/or year.

Cancer Statistics Data Format

Your file must be a .csv with the following structure:

- The first column must be titled City (case-insensitive)
- The second column must be titled State (case-insensitive)
- All subsequent columns must be either cancer type names (alphabetical text only) or years (numeric digits only)
- Each row represents one city
- State values must be valid 2-letter US state abbreviations (e.g., RI, MA, NY)

Example format:

```
City,State,Bladder,Breast,Lung,Prostate,2015,2016,2017,2018,2019,2020,2021
Providence,RI,11,50,40,30,47,45,44,42,41,39,37
Lincoln,RI,32,20,15,10,18,35,16,15,14,13,12
Barrington,RI,33,25,18,12,23,55,21,20,19,18,17
```

Important formatting rules for Cancer Statistics:

Rule	Detail
Minimum columns	At least 3 columns required: City, State, and one data column
Column names	Cancer type columns must contain only letters (e.g., Lung, Breast). Year columns must contain only digits (e.g., 2020).
State abbreviation	Must be a valid 2-letter US state code. Full state names are not accepted.
Column order	City must be column 1 and State must be column 2. All other columns may appear in any order after that.
Data values	Numeric incidence counts per city. Blank or non-numeric cells may cause errors.

2. County Race/Ethnicity Data Format

This optional file adds demographic context to your visualization by supplying racial and ethnic population counts at the county level.

- The first column must be titled exactly County (case-sensitive)
- Subsequent columns can be any race or ethnicity categories you are tracking
- Values should be whole number population counts

Example format:

```
County,White,Hispanic or Latino,Black,American  
Indian,Asian,Pacific Islander,Other,Two or More  
Bristol,45355,1943,773,105,1285,1,500,2774  
Kent,147106,9665,3220,532,4882,40,3860,10723  
Newport,72063,5592,2840,362,1570,64,2533,6211  
Providence,402194,160323,53803,5362,28614,380,94730,75658  
Washington,116202,4578,1532,1024,2610,51,1992,6428
```

3. Age and Sex Population Data Format

This optional file provides a demographic breakdown of the population by sex and age group, which can be correlated with cancer incidence trends.

- The first column must be titled exactly Sex (case-sensitive)
- Subsequent columns represent age ranges, labeled as numeric ranges (e.g., 0-9, 10-19, 80+)
- Rows should represent Male and Female populations
- Values should be whole number population counts

Example format:

```
Sex,0-9,10-19,20-29,30-39,40-49,50-59,60-69,70-79,80+  
Male,55897,68369,78277,74753,63778,73199,67452,38527,16969  
Female,54344,66057,75425,72340,62843,75137,73676,46375,30832
```

Generating a Visualization

Once your files are uploaded, click the Generate Visualization button at the bottom of the Import Data page. The application will validate that at least one file has been successfully uploaded, then process your data and redirect you to an interactive custom cancer map.

If no files have been uploaded, an alert will prompt you to upload data first. If there is an error during processing, a message will describe the issue so you can review and correct your data before trying again.

Common Upload Errors

Error Message	Cause	Fix
First two columns must be "City" and "State"	Column headers are missing or in the wrong order	Ensure City is column 1 and State is column 2 in your cancer CSV
State column must contain valid 2-letter US state abbreviations	State values are full names, misspelled, or non-US codes	Replace all state values with standard 2-letter codes (e.g., RI, MA)
Column must be either a cancer type (text) or year (number)	A column name contains mixed characters like "Type1" or "2020-21"	Rename columns to be purely alphabetical or purely numeric
First column must be "County"	County Race file is missing the County header	Rename the first column to County exactly
First column must be "Sex"	Age/Sex file is missing the Sex header	Rename the first column to Sex exactly
Invalid file format. Please upload a CSV file.	A non-CSV file was selected	Save your data as a .csv file before uploading